MILITARY SPECIFICATION SHEET

BOLT, 100° REDUCED FLUSH HEAD, LOW INTERFERENCE, STRAIGHT SHANK, EXPANDABLE ALUMINUM ALLOY SLEEVE, ALLOY STEEL PIN, 108 KSI Fsu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bolt described herein shall consist of this document and the latest issue of MIL-B-8831.

1. Application criteria. The bolts shall be used in connections to provide a precision net fit or controlled interference fit. When selection provides for compatibility of size, length, materials and fit, the bolts shall be used in accordance with the limitations of MIL-B-8831/1. The bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-8985/3 or MIL-N-8922/1 shall be used with these bolts.

3. Oversize bolts. Second and third oversize bolts may replace MIL-B-8831/4 bolts for second and third repair operations.

4. Tooling. Wrenches shall be in accordance with MIL-W-8982.

5. Military installation. Installation shall be in accordance with Navy NAVWEPS 01-1A-8 and Air Force T.O. 1-1A-8.

6. Materials.

6.1 Pin. Pin material shall be alloy steel in accordance with MIL-S-6049 (AISI 8740) or MIL-S-8844 (AISI 4340).

6.2 Sleeve. Sleeve material shall be in accordance with QQ-A-250/4 or WW-T-700/3 for aluminum alloys 2024-T3 and 2024-T42. Sleeve material shall be in accordance with QQ-A-225/6 or QQ-A-200/3 for aluminum alloy 2024-T351.

7. Heat treatment.

7.1 Pin. The pin shall be heat treated in accordance with MIL-H-6875 and Rockwell hardness shall be 39-43.

7.2 Sleeves. Aluminum alloy 2024-T62 sleeves shall be heat treated in accordance with QQ-A-250/5.

FSC 5306

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8. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI B46.1.

9. Coatings or surface treatment.

9.1 Pin. The pin shall be aluminum coated in accordance with Kalgard 2240C, Type I, Class A or the equivalent.

9.2 Sleeve. The inside diameter of the sleeve shall be anodized in accordance with MIL-A-8625, Type 1, Class 1. The outside diameter shall not be coated.

10. Lubricants. The bolt outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132. Dry film lubricant shall be in accordance with MIL-L-8937 with 0.0001-inch overspray on the head and threads.

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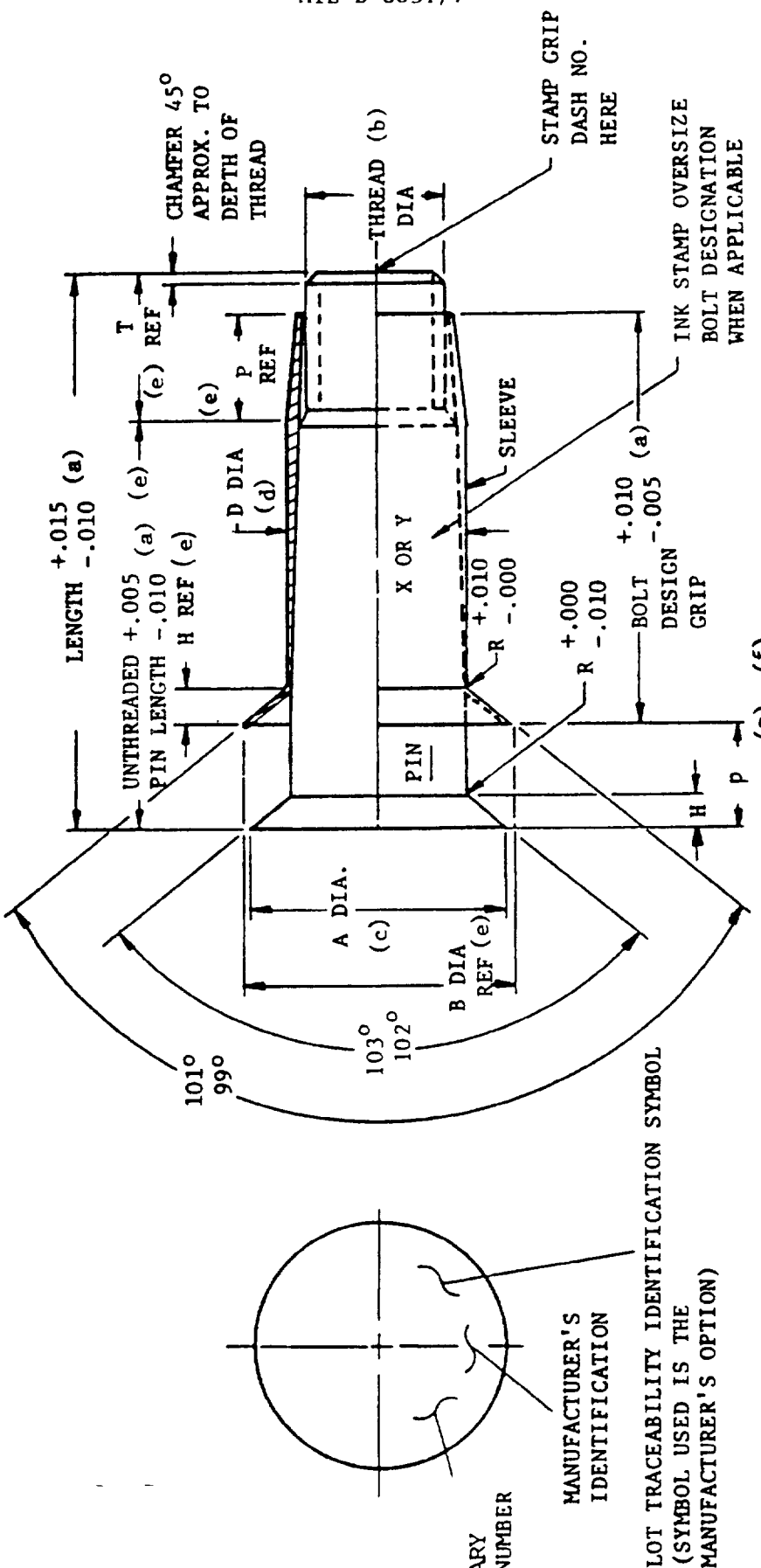


Figure 1. Reduced flush head bolt.

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NOTES:

a. Grip lengths are available in .0625 increments. See MIL-B-8831/1, Table I for bolt and grip lengths.

b. Threads shall be in accordance with MIL-S-8879 except thread major diameter (T.D.), which shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 T.I.R.

2. For sizes .437 thru .625, .012 T.I.R.

3. For sizes .750 thru 1.000, .020 T.I.R.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 T.I.R.

2. For sizes .437 thru .625, .010 T.I.R.

3. For sizes .750 thru 1.000, .016 T.I.R.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. For grip numbers -6 thry -8, taper ratios (reference only) are 25:1 for group I and 48:1 for groups II and III (see Table I).

g. Protrusion will vary according to taper to provide expansion as given in Table I.

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TABLE I. Bolt dimensions.

SIZE DASH NO's	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A BOLT HEAD	B REF SLEEVE FLANGE DIAMETER			M	FH FLANGE HEIGHT REF	P (g)	R FILLET RADIUS	T D THREAD DIA	T THREAD LENGTH	D DIMENSION (a) UNINSTALLED SLEEVE DIAMETER			D' DIMENSION (a) (b) EXPANDED SLEEVE DIAMETER			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
					STD	X O/S	Y O/S							STD	X	Y	STD	X	Y	STD	X O/S	Y O/S
-3	1	-6 THRU -8	1900-32	.3327 .3259	.3820	.3880	.3940	.053 .051	.061	.113 .103	.030	.1840 .1810	.343	2365 2360	2425 2420	2485 2480	2410 2400	2470 2460	2530 2520	.237	.243	.249
	2	-9 THRU -16		.3341 .3273	.3838	.3898	.3958		.062	.216 .206				2365 2360	2425 2420	2485 2480	2410 2400	2470 2460	2530 2520	.237	.243	.249
	3	-17 THRU -24		.3445 .3377	.4004	.4074	.4144		.064	.216 .206				2485 2480	2555 2550	2625 2620	2530 2520	2600 2590	.2670 2660	.249	.256	.263
-4	1	-6 THRU -8	2500-28	.4295 .4224	.4772	.4882	.4942	.068 .066	.074	.125 .115	.030	.2440 .2410	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3055 .3045	.3165 .3155	.3225 .3215	.301	.312	.318
	2	-9 THRU -17		.4339 .4268	.4805	.4915	.4975		.075	.240 .230				.3005 .3000	.3115 .3110	.3175 .3170	.3055 .3045	.3165 .3155	.3225 .3215	.301	.312	.318
	3	-18 THRU -28		.4469 .4398	.5011	.5081	.5171		.078	.240 .230				.3145 .3140	.3215 .3210	.3305 .3300	.3195 .3185	.3265 .3255	.3355 .3345	.315	.322	.331
-5	1	-6 THRU -8	3125-24	.5080 .5013	.5556	.5626	.5696	.074 .072	.080	.138 .128	.040	.3060 .3020	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3720 .3710	.3790 .3780	.3860 .3850	.367	.374	.381
	2	-9 THRU -20		.5163 .5098	.5630	.5700	.5770		.082	.264 .254				.3665 .3660	.3735 .3730	.3805 .3800	.3720 .3710	.3790 .3780	.3860 .3850	.367	.374	.381
	3	-21 THRU -31		.5306 .5233	.5836	.5916	.5986		.086	.264 .254				.3805 .3800	.3885 .3880	.3955 .3950	.3860 .3850	.3940 .3930	.4010 .4000	.381	.389	.396
-6	1	-6 THRU -8	3750-24	.5977 .5876	.6434	.6514	.6634	.084 .081	.090	.163 .153	.040	.3680 .3640	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4380 .4370	.4460 .4450	.4580 .4570	.432	.440	.452
	2	-9 THRU -22		.6087 .5986	.6536	.6616	.6736		.093	.312 .302				.4315 .4310	.4395 .4390	.4515 .4510	.4380 .4370	.4460 .4450	.4580 .4570	.432	.440	.452
	3	-23 THRU -34		.6243 .6142	.6754	.6834	.6914		.096	.312 .302				.4475 .4470	.4555 .4550	.4635 .4630	.4540 .4530	.4620 .4610	.4700 .4690	.448	.456	.464
-7	1	-6 THRU -8	4375-20	.7072 .6968	.7571	.7691	.7771	.103 .100	.108	.175 .165	.050	.4310 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5055 .5045	.5175 .5165	.5255 .5245	.499	.511	.519
	2	-9 THRU -26		.7234 .7130	.7729	.7849	.7929		.115	.336 .326				.4985 .4980	.5105 .5100	.5185 .5180	.5055 .5045	.5175 .5165	.5255 .5245	.499	.511	.519
	3	-27 THRU -40		.7416 .7312	.7974	.8084	.8204		.117	.336 .326				.5185 .5180	.5295 .5290	.5415 .5410	.5255 .5245	.5363 .5355	.5485 .5475	.519	.530	.542
-8	1	-6 THRU -8	5000-20	.7937 .7831	.8425	.8525	.8625	.112 .109	.114	.188 .178	.050	.4930 .4880	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5770 .5760	.5870 .5860	.5970 .5960	.570	.580	.590
	2	-9 THRU -30		.8157 .8051	.8637	.8737	.8837		.123	.360 .350				.5695 .5690	.5795 .5790	.5895 .5890	.5770 .5760	.5870 .5860	.5970 .5960	.570	.580	.590
	3	-31 THRU -46		.8366 .8260	.8908	.9008	.9098		.127	.360 .350				.5895 .5890	.5995 .5990	.6085 .6080	.5970 .5960	.6070 .6060	.6160 .6150	.590	.600	.609

NOTES

(a) "D" DIA SHALL BE VERIFIED WITHIN ITS GRIP BEARING PORTION BEFORE EXPANSION; WHEN SLEEVE IS PUSHED TO PIN HEAD, THE "D" DIA SHALL BE WITHIN THE EXPANDED "D" DIA LIMITS

(b) THIS BOLT IS MANUFACTURED AS A PRECISION ASSEMBLY REMOVAL OF SLEEVE FROM PIN BY USER WILL RESULT IN LOSS OF THE BOLT'S ABILITY TO MEET MIN/MAX EXPANSION LIMITS

(c) ALL DIMENSIONS ARE IN INCHES TOLERANCES: DECIMALS ± 010, ANGLES ±2°.

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TABLE II. Bolt weights.

GRIP NUMBER	BOLT DESIGN GRIP	-3 DIAMETER				-4 DIAMETER				-5 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
6	.375	.405	.748	.6037	.6436	405	.811	1.1145	1.1737	.405	.889	1.8481	1.9304
7	.438	.468	811	.6560	.7039	468	.874	1.2032	1.2738	.468	.952	1.9845	2.0821
8	.500	.530	.873	.7061	7625	530	.936	1.2888	1.3714	530	1.014	2.1166	2.2303
9	.562	.592	.935	.7874	.8417	.592	.998	1.4242	1.5019	.592	1.076	2.3440	2.4404
10	.625	.655	.998	.8424	.9037	655	1.061	1.5173	1.6049	.655	1.139	2.4893	2.5979
11	.688	.718	1.061	.8967	.9656	718	1.124	1.6094	1.7072	718	1.202	2.6334	2.7546
12	.750	.780	1.123	.9493	1.0256	.780	1.186	1.6992	1.8076	.780	1.264	2.7740	2.9083
13	.812	.842	1.185	1.0014	1.0855	842	1.248	1.7880	1.9073	.842	1.326	2.9135	3.0612
14	.875	.905	1.248	1.0053	1.0976	.905	1.311	1.8775	2.0080	.905	1.389	3.0542	3.2158
15	.938	.968	1.311	1.1051	1.2057	968	1.374	1.9661	2.1081	.968	1.452	3.1939	3.3698
16	1.000	1.030	1 373	1.1551	1.2643	1.030	1.436	2.0523	2.2062	1.030	1.514	3.3300	3.5206
17	1.062	1.092	1.435	1.3065	1.4370	1.092	1.498	2.1376	2.3037	1.092	1.576	3.4651	3.6708
18	1 125	1.155	1.498	1.3615	1.5020	1 155	1.561	2.4174	2.6029	1.155	1.639	3.6014	3.8227
19	1 188	1.218	1.561	1.4158	1.5664	1.218	1.624	2.5124	2.7113	1.218	1.702	3.7366	3.9738
20	1.250	1.280	1.623	1.4686	1.6296	1 280	1.686	2.6049	2.8175	1.280	1.764	3.8684	4.1220
21	1.312	1.234	1.685	1.5205	1.6923	1.234	1.748	2.6964	2.9231	1.342	1.826	4.2775	4.5557
22	1.375	1.405	1.748	1 5727	1.7555	1.405	1.811	2.7887	3.0298	1.405	1.889	4.4217	4.7177
23	1 438	1.468	1.811	1 6243	1.8183	1 468	1.874	2.8801	3.1359	1.468	1.952	4.5648	4.8789
24	1.500	1.530	1.873	1.6736	1.8791	1.530	1.936	2.9690	3.2399	1.530	2.014	4.7044	5.0371
25	1 562					1.592	1.998	3.0570	3.3433	1.592	2.076	4.8428	5.1944
26	1.625					1 655	2.061	3.1457	3.4478	1.655	2.139	4.9826	5.3536
27	1.688					1 718	2.124	3.2335	3.5517	1.718	2.702	5.1212	5.5120
28	1.750					1 780	2.186	3.3188	3.6534	1.780	2.264	5.2563	5.6674
29	1.812									1 842	2.326	5.3903	5.8220
30	1.875									1.905	2.389	5.5256	5.9783
31	1.938									1.968	2.452	5.6598	6.1340

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TABLE II. Bolt weights. - Continued

GRIP NUMBER	BOLT DESIGN GRIP	-6 DIAMETER				-7 DIAMETER				-8 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
6	375	405	.967	2.8819	2.9885	405	1 030	4.1857	4 3246	.405	1.108	5.8752	6 0684
7	438	468	1 030	3.0763	3.2023	468	1 093	4 4494	4.6127	.468	1.171	6.2174	6 4432
8	500	530	1.092	3.3269	3.4732	530	1.155	4.7058	4 8945	.530	1.233	6.5508	6.8105
9	562	592	1 154	3.5951	3.7117	592	1.217	5 1920	5.3238	.592	1.295	7.2196	7.3928
10	625	.655	1.217	3.8027	3.9339	655	1 280	5 4775	5 6257	.655	1.358	7.5940	7.7879
11	688	718	1.280	4.0090	4 1554	718	1 343	5 7615	5 9303	.718	1.421	7 9665	8.1812
12	750	780	1 342	4.2106	4 3726	780	1 405	6.0392	6 2221	.780	1.483	8.3312	5 5681
13	812	842	1.404	4.4108	4.5889	842	1 467	6.3153	6 5164	.842	1.545	8.6941	8.9539
14	875	905	1.467	4 6130	4.8078	905	1 530	6.5945	6 8144	.905	1.608	9.0611	9.3445
15	.938	968	1 530	4.8139	5.0258	968	1 593	6 8721	7 1114	.968	1.671	9.4264	9.7340
16	1 000	1 030	1 592	5.0102	5 2398	1 030	1 655	7 1426	7 4018	1.030	1.733	9.7839	10 1164
17	1 062	1 092	1 654	5.2042	5.4519	1 092	1 717	7 4125	7 6923	1.092	1.795	10.1396	10.4977
18	1.125	1 155	1 717	5.4020	5 6683	1 155	1 780	7 6853	7 9862	1.155	1.858	10 4995	10.8839
19	1.188	1 218	1 780	5.5966	5.8870	1.218	1.843	7.9566	8.2792	1.218	1 921	10.8576	11 2689
20	1.250	1 280	1,842	5.7875	6.0925	1 280	1 905	8 2218	8 5667	1.280	1.983	11.2080	11.6468
21	1 312	1 342	1 904	5.9771	6.3022	1 342	1 967	8.4854	8 8532	1 342	2.045	11 5552	12.0223
22	1 375	1 405	1.967	6 1687	6.5144	1.405	2 030	8.7521	9.1433	1.405	2 108	11.9079	12 4038
23	1 438	1 468	2.030	6 7516	7.1364	1.468	2.093	9.0171	9 4323	1 468	2.171	12.2588	12 7843
24	1 500	1 530	2.092	6 9531	7 3604	1 530	2 155	9.2762	9 7160	1.530	2.233	12.6021	13.1578
25	1 562	1 592	2.154	7 1531	7 5835	1.592	2 217	9 5338	9.9988	1 592	2.295	12.9436	13 5302
26	1 625	1 655	2.217	7 3554	7.8093	1 655	2.280	9.7943	10 2850	1.655	2.358	13.2892	13 9073
27	1 688	1 718	2 280	7.5563	8 0343	1.718	2 343	10 6798	11.2420	1.718	2.421	13.6331	14.2833
28	1 750	1 780	2 342	7.7524	8 2549	1 780	2 405	10 9542	11.5452	1.780	2.483	13.9694	14.6524
29	1 812	1 842	2 404	7 9472	8 4747	1 842	2 467	11 2271	11.8475	1.842	2 545	14.3040	15.0205
30	1 875	1 905	2 467	8.1441	8.6972	1.905	2.530	11.5031	12 1535	1.905	2 608	14 6426	15.3932
31	1 938	1 968	2 530	8.3397	8 9188	1 968	2 593	11 7776	12 4585	1.968	2.671	15.9146	16.7148
32	2 000	2 030	2 592	8.5306	9 1362	2 030	2 655	12 0458	12 7578	2 030	2.733	16.2719	17 1084
33	2 062	2 092	2 654	8 7201	9 3527	2 092	2 717	12.3125	13 0562	2 092	2 795	16 6274	17.5008
34	2 125	2.155	2 717	8 9117	9.5717	2 155	2 780	12.5823	13.3592	2.155	2.858	16 9874	17.8985
35	2 188					2 218	2 843	12.8505	13 6593	2.218	2.921	17.3455	18 2948
36	2.250					2 280	2.905	13.1126	13.9549	2.280	2.983	17.6957	18.6840
37	2 312					2 342	2.967	13 3731	14.2467	2.342	3.045	18.0441	19.0720
38	2 375					2.405	3.030	13 6367	14.5475	2.405	3.108	18.3969	19.4650
39	2 438					2.468	3.093	13.8988	14 8448	2.468	3.171	18.7479	19.8570
40	2.500					2.530	3 155	14.1548	15 1365	2.530	3.233	19.0911	20.2417
41	2.562									2.592	3.295	19.4325	20.6254
42	2.625									2.655	3.358	19.7782	21 0139
43	2.688									2.718	3 421	20.1722	21 4015
44	2 750									2.780	3.483	20.4584	21.7819
45	2 812									2.842	3 545	20 7928	22.1611
46	2 875									2 905	3 608	21 1316	22 5454

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TABLE III. Mechanical test requirements.

Diameter Dash Number	Ultimate Tensile Strength Lbs-Min	Double Shear Strength Lbs-Min (a)	Double Shear Nominal Test Dia. (a)	Pin Test Dia. Area (a)	Max. Install. Torque Inc Lbs
-3	2,290	6,100	.1900	.0284	40
-4	4,250	10,600	.2500	.0491	85
-5	5,950	16,600	.3125	.0767	140
-6	8,420	23,900	.3750	.1104	200
-7	11,600	32,500	.4375	.1503	300
-8	15,200	42,400	.5000	.1963	455

NOTES:

a. Shear test performed on pins ground to these diameters. The shear values are based on a minimum single shear strength of 108 KSI and the stress area at nominal test diameter.

b. Tensile fatigue and stress durability tests not applicable to 100° reduced flush head alloy.

11. Example of military part number.

M8831/4 - 6 X 34

- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- X — Oversize shank diameter: (X) = 1st oversize, (Y) = 2nd oversize. (Stamp on sleeve.)
- 6 — Diameter dash number in sixteenths of an inch.
- M8831/4 — Basic specification sheet number.

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12. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

b. Definitions of terms used in this document are in accordance with MIL-STD-1515.

c. For additional specification sheets and standards, see MIL-B-8831, Supplement 1.

d. The bolt assemblies identified herein are proprietary products of P.B. Fasteners and Deutsch Fastener Corporation and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997); 4,102,036 (expiring July 25, 1997); 4,048,898 (expiring September 20, 1996); Great Britain 1,524,886; Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. Additional patents are pending in France and Switzerland. The Government does not have a royalty-free license.

Custodians:
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Preparing activity
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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-B-8831/4 Bolt, 100° Reduced Flush Head, Low Interference, Double [illegible] Expandable Aluminum Alloy Sleeve, Alloy Steel Pin, 100 [illegible]

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN HELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

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